Modeling the Spread of Misinformation

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Abstract

This paper builds a compartmental ODE model for how a false rumor spreads through a population. The model adapts three compartments: ignorants, spreaders, and stiflers, and adds a debunking rate representing platform intervention. The Jacobian produces two stable policy thresholds: a minimum platform debunking rate $\delta^* = \beta N - \alpha$ and a minimum pre-emptive awareness campaign coverage $p^* = 1 - 1/R_0$. A suggested time-delayed extension later incorporates a deliberation lag τ representing the time between exposure and the decision to share. Simulations show that delay increases total exposure even when the equilibrium remains stable. The output of the model is three numbers that answer the question of how much intervention is needed and in what form.

1 Introduction

Misinformation spreads fast and causes real damage before it dies out. The World Health Organization has called the current information environment an infodemic, where the volume of false and misleading content makes it hard for accurate information to get through [5]. Trafficking in people and recognition ability is an example of where this becomes impactful. Many victims of human trafficking are exploited by someone they know, are disproportionately people of color and immigrants, and do not self-identify as trafficked. Starting in 2017, QAnon ran a large-scale, global campaign built around a completely different story: elite conspiracies, stranger kidnappings, and white children. By 2020, roughly 50 percent of Americans had seen QAnon content and about 14 percent identified as believers [3]. The result can be structural damage: survivors whose experiences do not match the narrative stop coming forward, providers get trained on the wrong profiles, real cases go unidentified.

This is a rumor spread problem and differential equations can model it. Daley and Kendall (DK) introduced the compartmental rumor model in 1964 by analogy with epidemic models [1], and Piqueira et al. validated this framework specifically for fake news on social networks in 2020 [4]. This paper utilizes the approach to derive intervention thresholds from the stability analysis and connects all results directly to platform moderation and community education strategies.

2 Part I: The ODE Model

2.1 Compartments and Assumptions

The population N is fixed over the timescale of a viral event, which is reasonable for a period of days to weeks. At any time $t \geq 0$, every person is in one of three groups. $I(t)$ is ignorants: people who have not heard the rumor. $S(t)$ is spreaders: people who have heard it and are actively sharing it. $R(t)$ is stiflers: people who have heard it but stopped sharing, either because they saw it debunked, lost interest, or noticed that everyone around them already knew it. The conservation law

$$I(t) + S(t) + R(t) = N$$

holds for all $t \geq 0$. People only move $I \to S \to R$.

The difference between this model and a standard disease SIR model, infected people recover on their own over time at a constant rate γ, giving a recovery term $-\gamma I$. Here a spreader stops sharing specifically because they contact someone who already knows the rumor, at which point it is no longer worth sharing. The stifling rate therefore scales with contact between spreaders and the already-informed population $S + R$, giving the term $-\alpha S(S + R)$. Using $I + S + R = N$ this becomes $-\alpha S(N - I)$. Stifling now is contact-driven, not time-driven, changing the equilibrium structure.

The following assumptions are made:

1. Population is homogeneously mixed: any person is equally likely to encounter any other person.
2. When an ignorant contacts a spreader, the ignorant becomes a spreader at per-contact rate $\beta > 0$.
3. When a spreader contacts any informed person, the spreader becomes a stifler at per-contact rate $\alpha > 0$.
4. In the extended model, platforms remove false content at per-capita rate $\delta \geq 0$, converting spreaders to stiflers independently of contact.

A compartment flow diagram is shown in Figure 1.

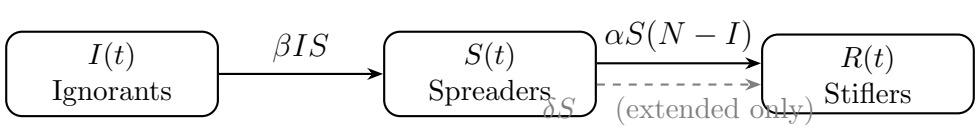

Figure 1: Compartment flow diagram

2.2 The ODE Systems

Each term below corresponds to one social interaction with a specific per-contact rate. The term βIS is contact between ignorants and spreaders. The term $\alpha S(N - I)$ is contact between spreaders and the informed population. The term δS is platform moderation acting independently of who talks to whom.

Baseline model ($\delta = 0$):

$$\begin{aligned}\dot{I} &= -\beta IS \\ \dot{S} &= \beta IS - \alpha S(N - I) \\ \dot{R} &= \alpha S(N - I)\end{aligned}$$

Extended model ($\delta > 0$):

$$\begin{aligned}\dot{I} &= -\beta IS \\ \dot{S} &= \beta IS - \alpha S(N - I) - \delta S \\ \dot{R} &= \alpha S(N - I) + \delta S\end{aligned}$$

Since $\dot{I} + \dot{S} + \dot{R} = 0$, the $\dot{R}$ equation is redundant. All analysis reduces to the two-dimensional system in I and S where $I \geq 0,\ S \geq 0,\ I + S \leq N$.

Parameters are defined:

Parameter	Description	Value Used
N	Fixed total population	100
β	Per-contact spreading rate	0.001
α	Per-contact stifling rate	0.01
δ	Per-capita platform debunking rate	0.012 (extended)

Table 1: ODE model parameters. $\alpha > \beta$ means people lose interest in sharing faster than they gain it, which is intentional and internally consistent.

2.3 Equilibria

Setting $\dot{I} = -\beta IS = 0$ gives $I = 0$ or $S = 0$. Setting $\dot{S} = 0$ with $S = 0$ gives $0 = 0$, for any I. Setting $I = 0$ in $\dot{S} = 0$ forces $S = 0$. Our equilibrium set is the line

$$\mathcal{E} = \big\{(I^*, 0) : 0 \leq I^* \leq N\big\}.$$

Every equilibrium has $S = 0$: the rumor always dies out. What matters is how much of the population is exposed before it does, which is $N - I^*$.

2.4 The Reproduction Number

R_0 is the expected number of new spreaders produced by one spreader dropped into a fully ignorant population. One spreader at $t = 0$ with $I(0) \approx N$ produces new spreaders at rate βN. The expected time spent as a spreader before stifling is $1/\alpha$. So:

$$R_0 = \frac{\beta N}{\alpha} = \frac{0.001 \times 100}{0.01} = 10.$$

If $R_0 > 1$ the rumor spreads. If $R_0 < 1$ it dies immediately.

2.5 The Critical Threshold I_c

Setting $\dot{S} = 0$ with $S \neq 0$ in the spreader equation and solving for I:

$$\beta I - \alpha(N - I) = 0 \implies (\alpha + \beta)I = \alpha N \implies I_c = \frac{\alpha N}{\alpha + \beta} = \frac{(0.01)(100)}{0.011} = 90.9.$$

When $I > I_c$, $\dot{S} > 0$ and spreaders grow. When $I < I_c$, $\dot{S} < 0$ and spreaders decline. Since $I(0) \approx N > I_c$, the rumor always grows at first, peaks when $I(t)$ hits $I_c = 90.9$, then declines.

2.6 Jacobian for Stability

The Jacobian is from $F(I, S) = -\beta IS$ and $G(I, S) = \beta IS - \alpha S(N - I)$:

$$J(I, S) = \begin{pmatrix} -\beta S & -\beta I \\ (\beta + \alpha)S & \beta I - \alpha(N - I) - \alpha S \end{pmatrix}.$$

Because R=N - I - S is always determined by the other two variables, the third equation does not carry new information so the system reduces to two dimensions. Then look at $(I^*, 0)$ by setting $S = 0$:

$$J^* = \begin{pmatrix} 0 & -\beta I^* \\ 0 & (\alpha + \beta)I^* - \alpha N \end{pmatrix}.$$

The eigenvalues are:

$$\lambda_1 = 0, \qquad \lambda_2 = (\alpha + \beta)I^* - \alpha N.$$

$\lambda_1 = 0$ says there is a continuum of equilibria along $S = 0$ not a single isolated fixed point. So stability is determinded by λ_2:

$$(\alpha + \beta)I^* < \alpha N \implies I^* < \frac{\alpha N}{\alpha + \beta} = I_c.$$

So $\lambda_2 < 0$ when $I^* < I_c$ and the rumor dies, and $\lambda_2 > 0$ when $I^* > I_c$ and the rumor spreads. This is where $I_c = 90.9$ is validated. Piqueira et al. (2020) apply the identical Jacobian approach to the DK fake-news model and validate the stability condition [4].

2.7 Decision-Making Thresholds

Next find the parameter value that pushes R_0 below 1 for each intervention strategy.

2.7.1 Minimum Platform Debunking Rate

Adding δ changes the denominator of R_0:

$$R_0^\delta = \frac{\beta N}{\alpha + \delta}.$$

Setting $R_0^\delta < 1$ and solving for δ:

$$\frac{\beta N}{\alpha + \delta} < 1 \implies \beta N < \alpha + \delta \implies \delta > \beta N - \alpha.$$

$$\boxed{\delta^* = \beta N - \alpha = (0.001)(100) - 0.01 = 0.09.}$$

βN is the spreading pressure and α is the natural stifling already in the system. Debunking only needs to close the gap between them.

2.7.2 Minimum Awareness Campaign Coverage

Pre-educating fraction p of the population shrinks the effective ignorant pool to $(1-p)N$:

$$R_0^{(p)} = \frac{\beta(1-p)N}{\alpha}.$$

Setting $R_0^{(p)} < 1$ and solving for p:

$$\frac{\beta(1-p)N}{\alpha} < 1 \implies 1 - p < \frac{\alpha}{\beta N} \implies p > 1 - \frac{1}{R_0}.$$

$$\boxed{p^* = 1 - \frac{1}{R_0} = 1 - \frac{1}{10} = 0.90.}$$

(As seen as the vaccination threshold in SIR.) For $R_0 = 10$, reaching 90 percent of the community in advance prevents the rumor from spreading at all.

3 Part II: The Time-Delayed Extension

3.1 Motivation and Model

The ODE model assumes that a person who hears a rumor immediately decides whether to spread it or not. However, people may see a post at any time and sit on it, then choose to

share it hours or days later. To capture this, a lag $\tau > 0$ is introduced, representing the time between first exposure and the spreading decision.

From [2], the delayed logistic model replaces the instantaneous density term $X(t)/K$ with the lagged version $X(t-\tau)/K$:

$$\frac{dX}{dt} = rX(t)\left(1 - \frac{X(t-\tau)}{K}\right).$$

But delayed feedback can cause the system to overcompensate and this can produce oscillations when τ is large relative to the response time $1/r$. Also, when people delay their sharing decision, the spreader inflow at time t reflects contact information from time $t-\tau$, not from right now.

$$\begin{aligned}
\dot{I}(t) &= -\beta I(t-\tau)\, S(t-\tau) \\
\dot{S}(t) &= \beta I(t-\tau)\, S(t-\tau) - \alpha S(t)(N - I(t)) \\
\dot{R}(t) &= \alpha S(t)(N - I(t))
\end{aligned}$$

The stifling term $\alpha S(t)(N - I(t))$ stays undelayed because going quiet is an immediate response: a spreader encounters someone who already knows the rumor and stops in that moment. Only the spreading decision is lagged. The new parameter introduced is simulated at time units 0, 10, 30.

3.2 Stability of the Delayed System

Next, test what happens when spreaders are given a small nudge away from zero by assuming $S(t) = e^{\lambda t}$ and plug it in. If $\lambda < 0$ the spreaders decay back to zero and the equilibrium is stable; if $\lambda > 0$ they grow and it is unstable. The complication is that the delay term $S(t-\tau)$ becomes $e^{\lambda t}e^{-\lambda\tau}$ after substitution, meaning λ now appears in the equation in two places so theres no clean formula for λ [2].

The simulation runs the delayed model forward from a small perturbation and check whether $S(t)$ decays or grows. Then the boundary between stable and unstable by setting $\lambda = i\omega$ is the point where the system tips from decaying to oscillating. Solving the resulting real and imaginary parts finds the critical delay τ^* where stability changes, if there is one. For small-τ approximation for small delays, $e^{-\lambda\tau} \approx 1 - \lambda\tau$, gets:

$$\lambda \approx \frac{(\alpha+\beta)I^* - \alpha N}{1 + \beta I^* \tau}.$$

The numerator is λ_2 from the Jacobian originally. When $I^* < I_c = 90.9$, the numerator is negative, so $\lambda < 0$ for any small τ as well. Small delays dont change the stability from the original system. When $I^* < I_c = 90.9$ that numerator is negative, so $\lambda < 0$ for any $\tau > 0$ as well. This means small delays do not destabilize the rumor-free equilibrium. The ODE stability carries over for small τ, and the threshold $I_c = 90.9$ still decides whether spreaders grow or decay.

4 Simulations

All simulations are implemented in Python using `numpy`, `scipy.integrate.solve_ivp` for the ODE systems, `scipy.optimize.fsolve` for root-finding on the characteristic equation, and `matplotlib` for all plots. The delayed model is solved using a forward Euler strategy with history tracking, where lagged values $I(t-\tau)$ and $S(t-\tau)$ are looked up from stored arrays at each time step, mirroring the approach in Barnes and Fulford [2].

Using $\beta = 0.001$, $\alpha = 0.01$, $N = 100$, $I(0) = 99$, $S(0) = 1$, $R(0) = 0$. Derived $R_0 = 10$, $I_c = 90.9$, $\delta^* = 0.09$, $p^* = 0.90$.

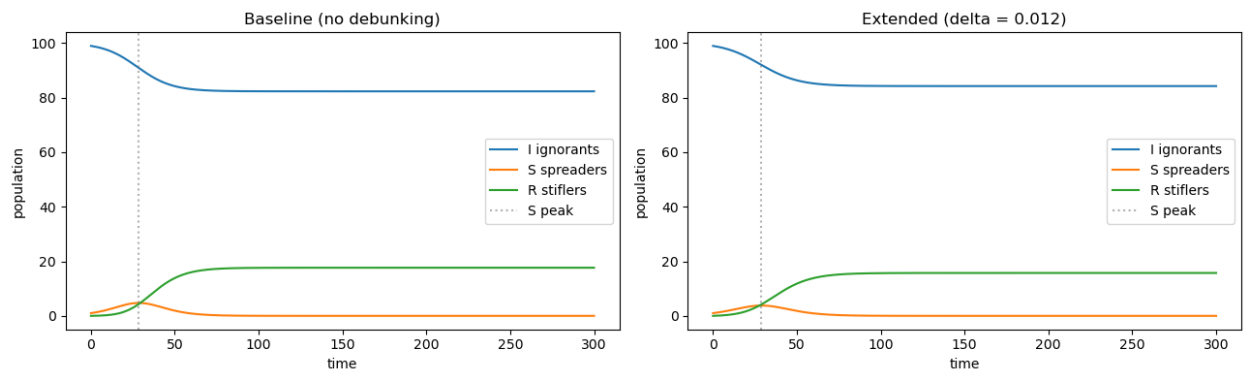

Figure 2: Time series. Without intervention, $I(t)$ falls from 99 to about 83, meaning 17 percent of the population is exposed. $S(t)$ peaks around $t = 30$ when I crosses the critical threshold $I_c = 90.9$ from the nullcline again. With $\delta = 0.012$, which is below the minimum debunking rate $\delta^* = 0.09$, exposure drops to 15 percent. Less than threshold debunking still helps.

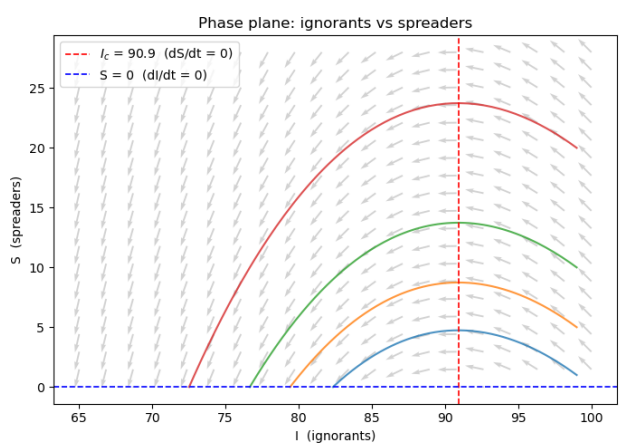

Figure 3: Phase plane. Trajectories are shown with $I(0) = 99$ and $S(0) \in \{1, 5, 10, 20\}$. Since $\dot{I} = -\beta IS < 0$ always everything moves and vector field point left, peaks at the nullcline $I = I_c = 90.9$ and falls to $S = 0$. The vector field is the visual version of the eigenvalue result $\lambda_2 > 0$ on the right, $\lambda_2 < 0$ on the left.

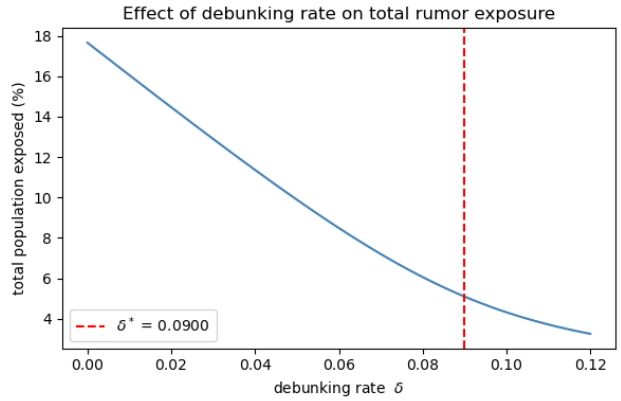

Figure 4: Decision boundary. At $\delta = 0$, 17.5 percent of the population is exposed. At the minimum debunking rate $\delta^* = 0.09$, exposure drops to about 4 percent. The curve is steep early, so the first increments of moderation do the most work. Reaching δ^* is not required to meaningfully reduce harm.

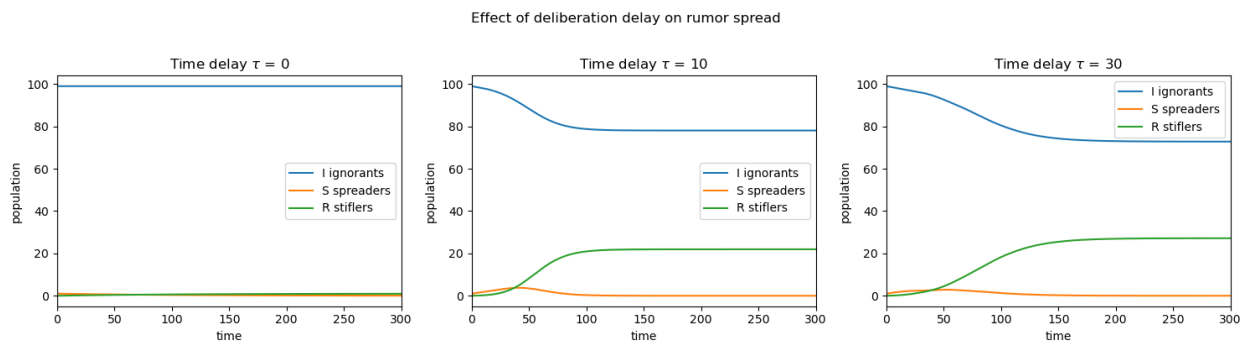

Figure 5: Time-delay simulations. In Python an Euler scheme with history tracking [2] proposes results for $\tau = 0$, $\tau = 10$, and $\tau = 30$.

The most immediate finding and difference is that delay increases total exposure: at $\tau = 0$, 1 percent of the population is reached; at $\tau = 10$, this rises to 21.9 percent; at $\tau = 30$, it reaches 27.2 percent. This is counterintuitive. A person deliberating longer before sharing does not reduce harm, it increases it. The reason is that the delayed spreading term $\beta I(t-\tau)S(t-\tau)$ feeds new spreaders based on stale contact information. The system loses real-time awareness of how much spreading pressure exists, so spreaders accumulate more slowly but persist longer before stifling catches up.

Solving for the roots found no imaginary axis crossing so the equilibrium stays stable for all τ. The rumor still burns out, just more slowly and with more total damage. Platforms that rely on organic stifling to kill a rumor cannot count on it acting quickly when users deliberate before sharing. Since stifling is instantaneous but new spreaders arrive late, the natural self-regulation weakens as τ grows. This makes active platform debunking at $\delta > \delta^* = 0.09$ more important, not less, in a world where deliberation delays are realistic.

5 Conclusions

Our model pushes platforms to need $\delta > 0.09$ to suppress a rumor from the start. Awareness organizations need to reach at least 90 percent of the target community before the rumor enters to prevent it from spreading at all. When people delay before sharing, total exposure increases from 1 percent at $\tau = 0$ to 27 percent at $\tau = 30$, so the natural self-regulation of the rumor weakens as the deliberation time to share window grows. Even below both thresholds though, moderation reduces total exposure.

For the trafficking context, these results support two concrete strategies. Platforms should maintain content moderation above $\delta^* = 0.09$ for known false trafficking narratives, the time-delay result making this more urgent. Platform-level intervention that acts in real time becomes the more reliable lever. Anti-trafficking organizations should treat pre-emptive community education as the proactive complement: reaching 90 percent of a community with accurate information about what trafficking actually looks like prevents the QAnon-style narrative from taking hold before it enters. This is consistent with the recommendations in Prakash et al. who call for health sector education to reach diverse demographics before disinformation saturates the space [3].

The main limitation is the homogeneous mixing assumption. Real social media has echo chambers where spreaders rarely encounter informed people, keeping the stifling term $\alpha S(N - I)$ small and letting the rumor persist longer than this model predicts. The time-delay extension compounds this: in a clustered network with delayed sharing behavior, both $\delta^* = 0.09$ and $p^* = 0.90$ would be higher than the values derived here. A network model where N is replaced by a local cluster/echo chamber size and τ is estimated from platform engagement data is a future step for making these thresholds more realistic.

References

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[2] B. Barnes and G. Fulford, *Mathematical Modeling with Case Studies*, 3rd ed. CRC Press, 2015.

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[5] World Health Organization, "Infodemic," 2022. Available at: `https://www.who.int/health-topics/infodemic`